Conventional Power Generation

Unit 3

Unit 3: Conventional Power Generation

Steam power station, Nuclear power plant – Gas turbine power plant- Hydro power station: Schematic arrangement, advantages and disadvantages, Thermo electric and thermionic generators, Environmental aspects for selecting the sites and locations of power plants.

What are conventional energy sources ?

The energy sources that once exhausted, do not replenish themselves within a specific period are called conventional or non -renewable energy sources.

Conventional energy sources includes oil, gas and coal. These conventional sources are usually fossil fuels. Their use leads to increased greenhouse gas emissions and other environmental damage These sources of energy are also called nonrenewable sources.

Benefits of conventional energy sources

- Large amount of power can be generated by using coal and nuclear etc.
- Transportation of oil and coal to the plant is easy.
- The efficiency of the energy sources is high.
- The production expenses are low.
- Generally it doesn’t need any specific place for installation.

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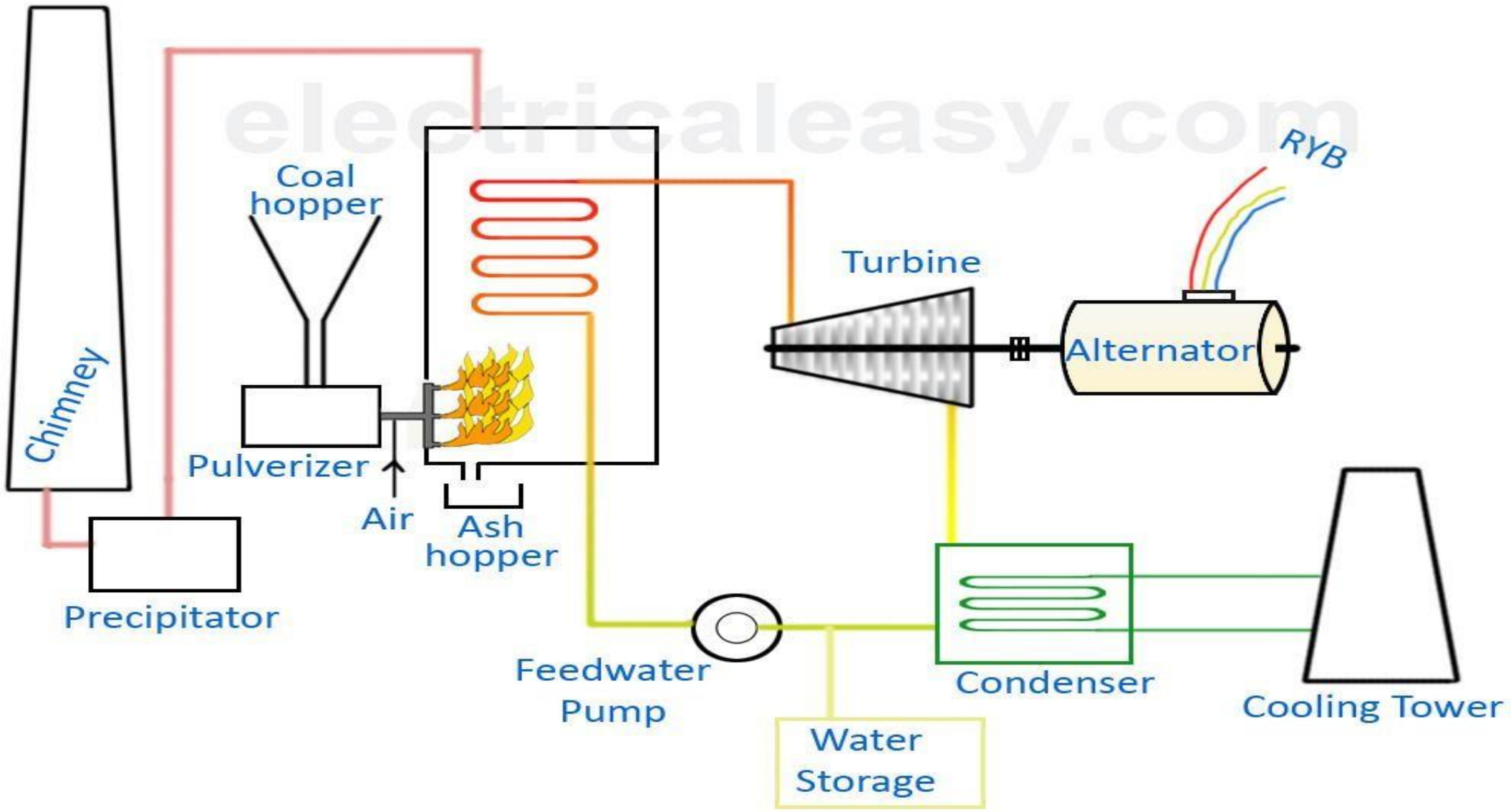
- **Fossil fuels used for generation of conventional power**

In steam or thermal power plant coal, oil or natural gas may be used as a fuel. In a fossil fuel power plant the chemical energy stored in fossil fuels such as coal, fuel oil, natural gas or oil shale and oxygen of the air is converted successively into thermal energy, mechanical energy and, finally, electrical energy. Fossil fuel power stations have machinery to convert the heat energy of combustion into mechanical energy, which then operates an electrical generator. The prime mover may be a steam turbine, a gas turbine or, in small plants, a reciprocating gas engine. All plants use the energy extracted from expanding gas, either steam or combustion gases.



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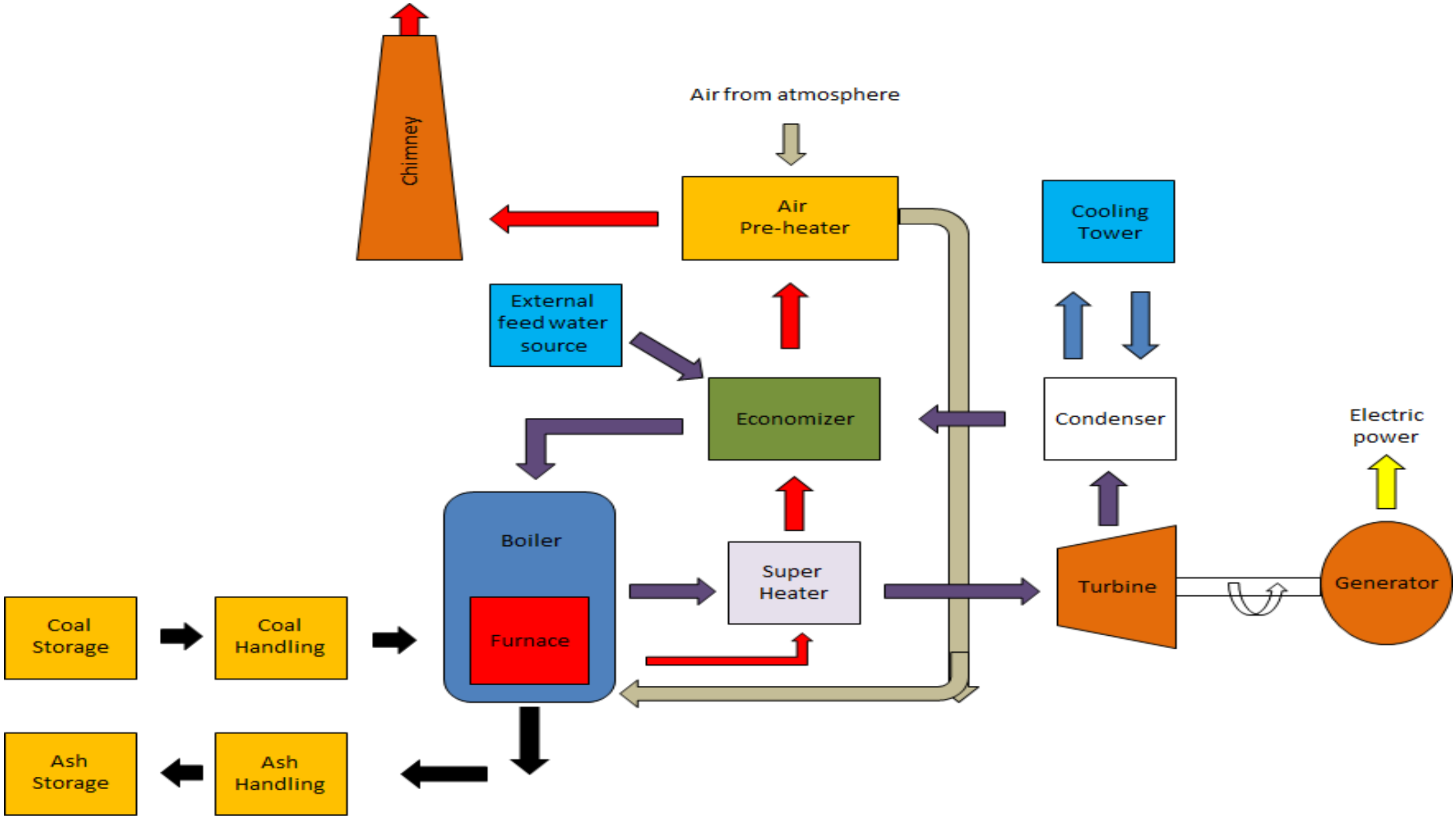
In coal thermal power plant, the steam is produced in high pressure in the steam boiler due to burning of fuel (pulverized coal) in boiler furnaces. This steam is further supper heated in a super heater. This supper heated steam then enters into the turbine and rotates the turbine blades. The turbine is mechanically so coupled with alternator that its rotor will rotate with the rotation of turbine blades. After entering in turbine the steam pressure suddenly falls and corresponding volume of the steam increases. After imparting energy to the turbine rotor the steam passes out of the turbine blades into the condenser. In the condenser the cold water is circulated with the help of pump which condenses the low pressure wet steam. This condensed water is further supplied to low pressure water heater where the low pressure steam increases the temperature of this feed water, it is again heated in high pressure.


Chimney
Air from atmosphere
Air
Pre-heater
Cooling
Tower
External
feed water
source
Economizer
Condenser
Electric
power
Boiler
Furnace
Super
Heater
Turbine
Generator
Coal
Storage
Coal
Handling
Ash
Storage
Ash
Handling

Components of steam power plants as follows:

- Coal handling plant
- Pulversing plant
- Ash handling plant/ Fly ash circuit
- Boiler
- Steam turbine and generator
- Condenser
- Cooling tower
- Feed pump
- Economizer
- Superheater
- Air preheater

- **Coal Storage:** It is the place where coal is stored which can be utilized when required.
- **Coal Handling:** Here the coal is converted into the pulverized form before feeding to the furnace. A proper system is designed to transport the pulverized coal to the boiler furnace.
- **Boiler:** The boiler is responsible for heating water to generate steam. This is typically achieved by burning fossil fuels such as coal, oil or natural gas.
- **Air-preheater:** It is used to pre-heat the air before entering into the boiler furnace. The pre heating of air helps in the burning of fuel to a greater extent. It takes the heat from the burnt gases from the furnace to heat the air from the atmosphere.
- **Economizer:** As its name indicates it economizes the working of the boiler. It heats the feed water to a specified temperature before it enters into the boiler drum. It takes the heat from the burnt gases from the furnace to do so.
- **Turbine:** It is the mechanical device which converts the kinetic energy of the steam to the mechanical energy.

- **Generator:** It is coupled with the turbine rotor and converts the mechanical energy of the turbine to the electrical energy.
- **Ash handling plant/ Fly ash circuit:** It is used to store the ash after burning of the coal. Coal contain considerable amount of ash. Percentage of ash vary from 5 to 40
- **Dust Collector:** It collects the dust particle from the burnt gases before it is released to the chimney.
- **Condenser:** It condensate the steam that leaves out turbine. It converts the low-pressure steam to water. It is attached to the cooling tower.
- **Cooling Tower**: It is a tower which contains cold water. Cold water is circulates to the condenser for the cooling of the residual steam from the turbine.
- **Chimney:** It is used to release the hot burnt gases or smoke from the furnace to the environment at appropriate height. The height of the tower is very high such that it can easily throw the smoke and exhaust gases at the appropriate height. And it cannot affect the population living near the steam power plant.
- **Feed Water pump:** It is used to transport the feed water to the boiler.

Working

- The pulverized coal is fed into the boiler where the pulverized coal is burnt into the furnace
- Due to heat from the furnace, the water present in the boiler drum changes to the high pressure steam.
- From the boiler this high-pressure steam is passed to the super heater where it is again heated up to its dryness.
- After that this super-heated steam strikes the turbine blades with a high speed and the turbine blades starts rotating to at high speed. Here the Steam Kinetic energy of the steam is gets converted into mechanical energy.
- A generator is coupled with the turbine rotor. As the turbine rotates, the generator also rotate with same speed and mechanical energy of the turbine gets converted into electrical energy.
- Steam after hitting the turbines blades lost its most of the energy and leaves the turbine with low pressure steam.

- This low-pressure steam enters into the condenser. Cold water circulates in the condenser from the cooling tower. Here the low-pressure wet steam is converted into water.
- After that condensed water with the feed water passed to the economizer where it gets heated up by the economizer. And finally, the feed water enters into the boiler by a feed water pump to repeat the cycle.
- The burnt flue gases from the furnace passes through the super heater, economizer and air pre-heater. This heat of the flue gases is used to heat the steam in the super heater to its dryness, to heat feed water in the economizer before entering into the boiler and to heat air form the atmosphere in the air pre-heater before it enters into the furnace.
- The ash from the furnace is transported to ash handling plant and finally to the ash storage.

Advantages of steam power generation:

- As compared with power generating plant, it has low initial cost and hence economical.
- Less land area is required as compared with the hydro power plant.
- Coal is used as fuel and the cost of coal is cheaper than petrol and diesel fuel. So the power generation cost is economical.
- This power plant has easy maintenance cost.
- Steam power plant can be installed in any area where water sources and transportation facility are easily available.

Disadvantages of steam power generation:

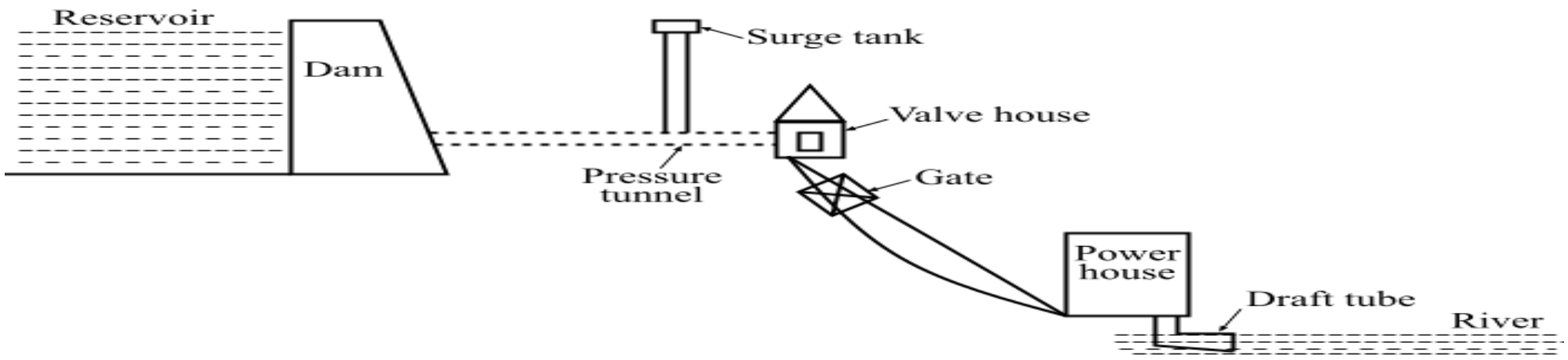
- The running cost of steam power plant is comparatively high because of fuel, maintenance etc.
- If we talk about the overall efficiency of steam power plant, than is about 35 % to 41% which is low.
- Due to the release of burnt gases of the coal or fuel, it contributes to the global warming to a larger extent.
- The heated water that is thrown in the rivers, ponds etc. puts and adverse effect on the living organism of water and disturbs the ecology.
- Time needed is large.

What are Site selection criteria for selecting location of steam power plant?

- Availability of raw material
- Availability of raw fuel
- Geology and soil type
- Land availability and its cost
- Load center
- Pollution impact
- Ash disposal facility
- Transport facility
- Distance from populated area.

Hydroelectric power plant

- A hydroelectric power plant is a generating station which converts the potential energy of water at high level into electrical energy.
- Generally, the hydroelectric power plants are installed in hilly areas where dams can be built, and large water reservoirs can be obtained. In a hydroelectric power plant, the water head is created by constructing a dam across a river. From the dam, water is led to a water turbine. The water turbine converts the hydraulic energy of the falling water into mechanical energy. The turbine drives the alternator, which converts the mechanical energy of the turbine into electrical energy.



Site Selection for Hydroelectric Power Plant

The following factors should be considered while selecting the site for a hydroelectric power plant −

Availability of Water – Since water is the main requirement for a hydroelectric power plant. Hence, the hydroelectric power plant should be built at a place a huge amount of water is available at a good head.

Storage of Water – The site selected for the hydroelectric power plant should provide adequate facilities for constructing a dam and storage of water to ensure the generation of power throughout the year.

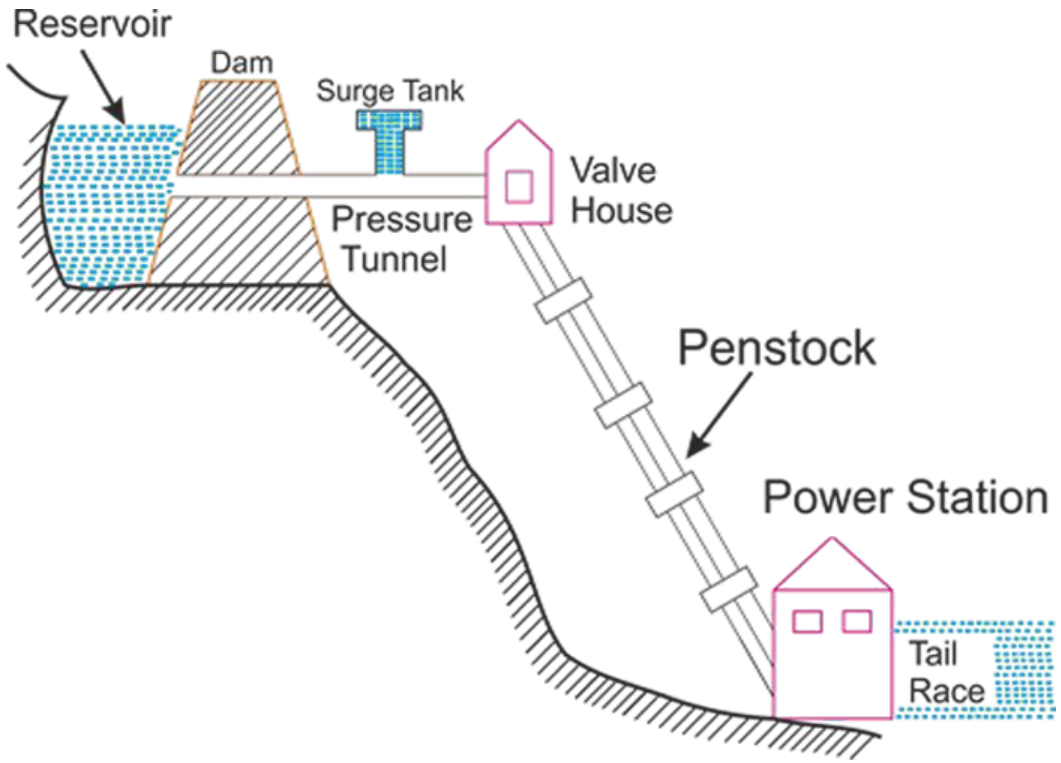
Type and Cost of Land – The bearing capacity of the land should be adequate so that heavy equipment can be installed. Also, the land for the construction of the hydroelectric power plant should be available at reasonable price.

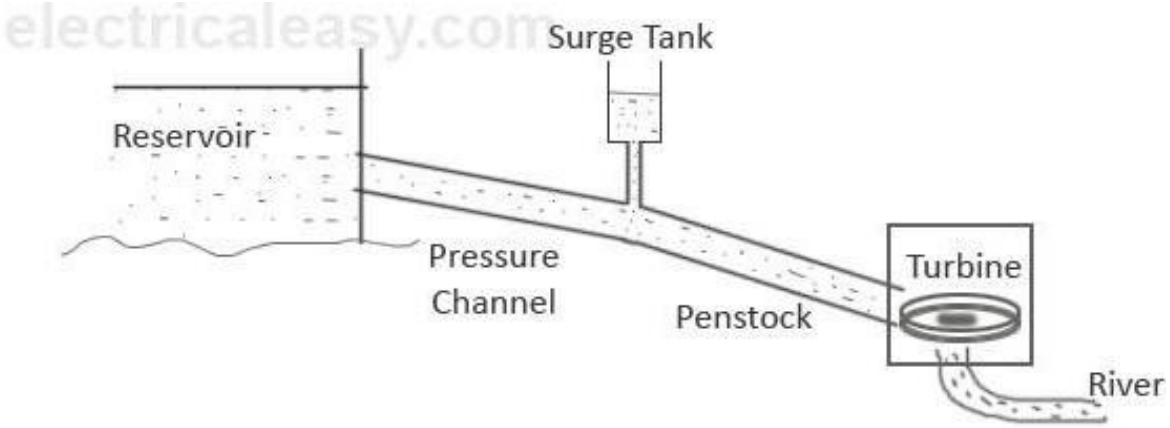
Transportation Facilities – A hydroelectric power plant requires transportation of equipment and machinery. Therefore, the site selected for the power plant should be well connected to other parts of the country by rail, road, etc.

COMPONENTS OF HYDRO ELECTRIC POWER PLANT

- **Dam and Reservoir**: The dam is constructed on a large river in hilly areas to ensure sufficient water storage at height. The dam forms a large reservoir behind it. The height of water level (called as water head) in the reservoir determines how much of potential energy is stored in it.
- **Control Gate:** Water from the reservoir is allowed to flow through the penstock to the turbine. The amount of water which is to be released in the penstock can be controlled by a control gate. When the control gate is fully opened, maximum amount of water is released through the penstock.
- **Penstock:** A penstock is a huge steel pipe which carries water from the reservoir to the turbine. Potential energy of the water is converted into kinetic energy as it flows down through the penstock due to gravity.
- **Water Turbine:** Water from the penstock is taken into the water turbine. The turbine is mechanically coupled to an electric generator. Kinetic energy of the water drives the turbine and consequently the generator gets driven. There are two main types of water turbine; (i) Impulse turbine and (ii) Reaction turbine.
- **Generator:** A generator is mounted in the power house and it is mechanically coupled to the turbine shaft. When the turbine blades are rotated, it drives the generator and electricity is generated which is then stepped up with the help of a transformer for the transmission purpose.

- **Surge Tank:** Surge tanks are usually provided in high or medium head power plants when considerably long penstock is required. A surge tank is a small reservoir or tank which is open at the top. It is fitted between the reservoir and the power house. The water level in the surge tank rises or falls to reduce the pressure swings in the penstock. When there is sudden reduction in load on the turbine, the governor closes the gates of the turbine to reduce the water flow. This causes pressure to increase abnormally in the penstock. This is prevented by using a surge tank, in which the water level rises to reduce the pressure. On the other hand, the surge tank provides excess water needed when the gates are suddenly opened to meet the increased load demand.





WORKING

- Initially the water of the river is in Catchments Area.
- From catchments area the water flows to the dam.
- At the dam the water gets accumulated . Thus the potential energy of the water increases due to the height of the dam .
- When the gates of the dam are opened then the water moves with high Kinetic Energy into the penstock.
- Through the penstock water goes to the turbine house.
- Since the penstock makes water to flow from high altitude to low altitude, Thus the Kinetic Energy of the water is again raised.

- In the turbine house the pressure of the water is controlled by the controlling valves as per the requirements.
- The controlled pressurized water is fed to the turbine.
- Due to the pressure of the water the light weight turbine rotates.
- Due to the high speed rotation of the turbine the shaft connected between the turbine and the generator rotates .
- Due to the rotation of generator the ac current is produced.
- This current is supplied to the powerhouse .
- From powerhouse it is supplied for the commercial purposes.

Types Of Hydro-Power Plants

1. Conventional Plants
2. Pumped Storage Plant:
3. Run-Of-River Plant:

Advantages Of A Hydroelectric Power Plant

- Neat and clean source of energy
- Serves other purposes too, such as irrigation
- Water is used for the generation of electrical energy. Therefore, it requires no fuel.
- No smoke and ash is produces, hence it is quite neat and clean.
- As water is the source of energy which is available free of cost. Thus, the hydroelectric power plant requires very small running cost.
- Hydroelectric power plant is simple in construction and requires less maintenance.
- The hydroelectric power plants can be put into service instantly, i.e. it does not require a long starting time.

- Apart from the generation of electrical energy, the hydroelectric power plants also help in irrigation and controlling floods

Disadvantages

- Very high capital cost due to construction of dam
- High cost of transmission – as hydro plants are located in hilly areas which are quite away from the consumers.
- For the construction of hydroelectric power plants, skilled and experienced personals are required.
- There is uncertainty about the availability of water because it depends upon the weather conditions.

Nuclear Reactor

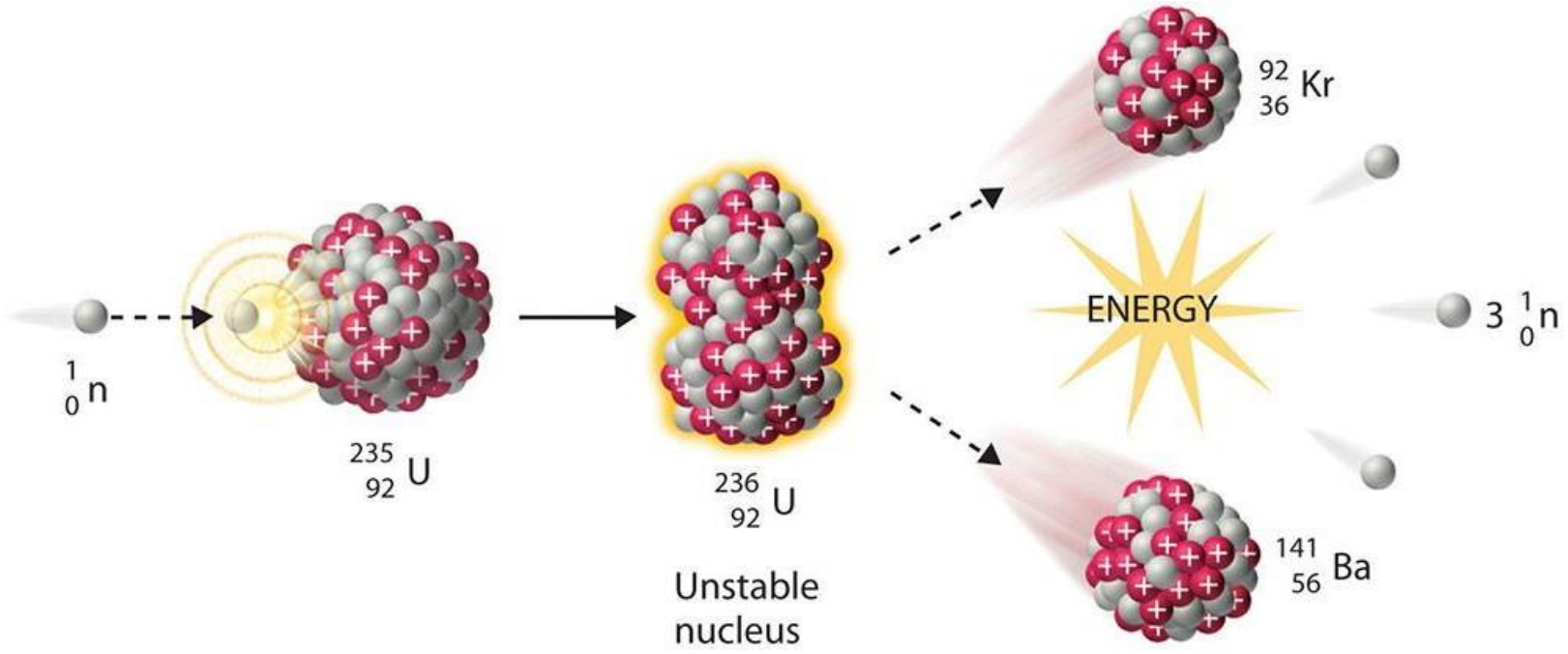
- **Nuclear Reactor :** A reactor is a device in which nuclear chain reactions are initiated, controlled, and sustained at a steady rate, as opposed to a nuclear bomb, in which the chain reaction occurs in a fraction of a second and is uncontrolled causing an explosion.

- Fuel – Heavy metal fuel like U_{235} is used. It is in the form of small, hard ceramic pellets are packaged into long vertical tubes. Bundles of this fuel are insulated into reactor.

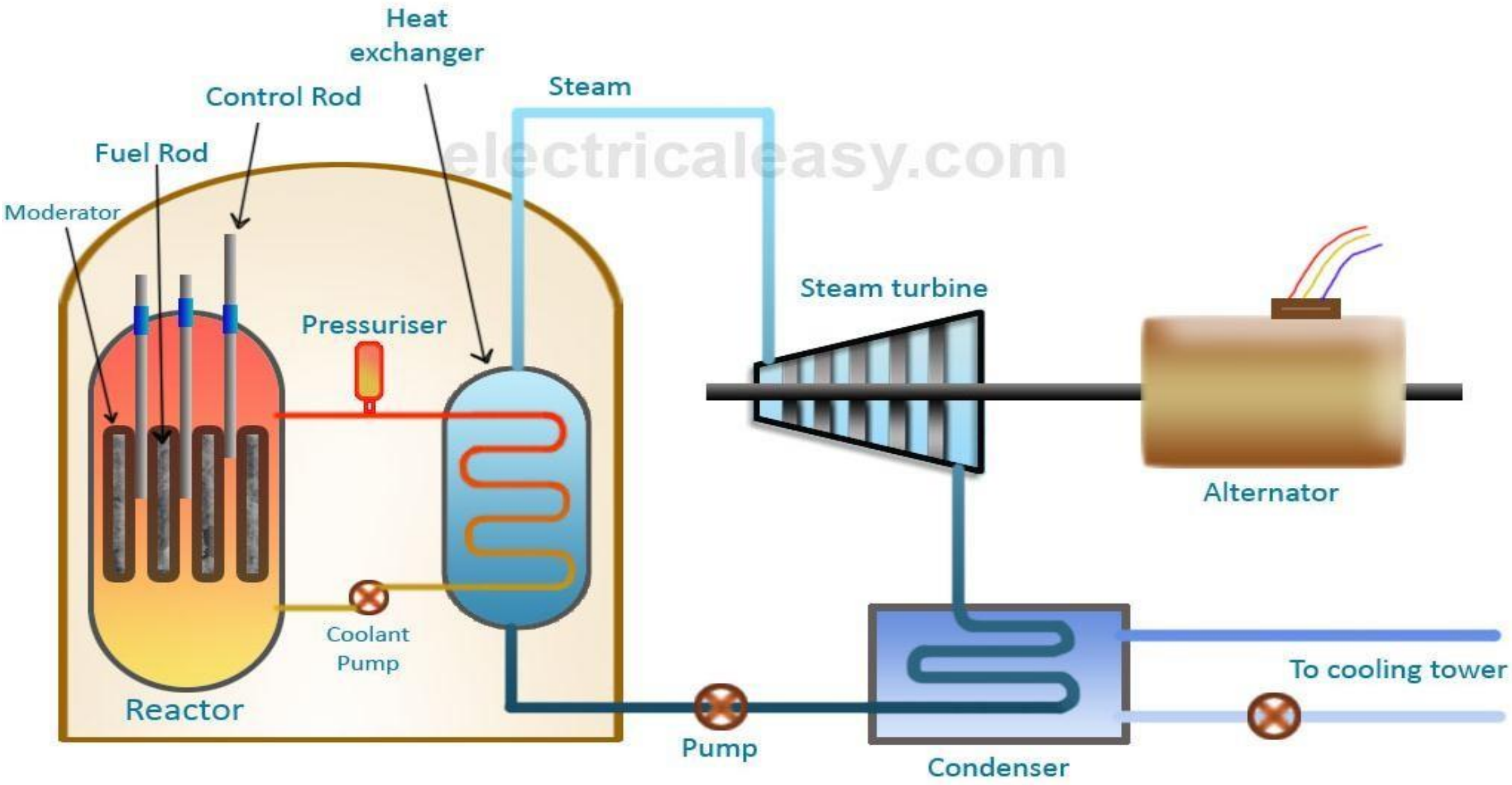
- In a **nuclear power plant**, heat energy is generated by a nuclear reaction called as **nuclear fission**. It is found that fission of only 1 Kg of Uranium produces as much heat energy as that can be produced by 4,500 tons of high-grade coal. This considerably reduces the transportation cost of fuel, which is a major **advantage of nuclear power plants**. Also, there are large deposits of nuclear fuels available all over the world and, hence, nuclear power plants can ensure continued supply of electrical energy for thousands of years. About 10% of the total electricity of the world is generated in nuclear power plants.

NUCLEAR CHAIN REACTIONS

- A chain reaction refers to a process in which neutrons released in fission produce an additional fission in at least one further nucleus. This nucleus in turn produces neutrons, and the process repeats. If the process is controlled, it is used for nuclear power or if uncontrolled it is used for nuclear weapons.
- $U^{235} + n \rightarrow$ fission + 2 or 3 n + 200 MeV
- If each neutron releases two more neutrons, then the number of fissions doubles each generation. In that case, in 10 generations there are 1,024 fissions and in 80 generations about 6×10^{23} (a mole) fissions.



- Nuclear Reactor



Component of Nuclear Reactor

- Fuel
- Nuclear reactor
- Moderator
- Turbine
- Control rods
- Shielding
- Steam Generator
- Coolant

Nuclear Reactor

Since the nuclear fission is radioactive, the reactor is covered by a protective shield. Splitting up of nuclei of heavy atoms is called as nuclear fission, during which huge amount of energy is released. Nuclear fission is done by bombarding slow moving neutrons on the nuclei of heavy element. As the nuclei break up, it releases energy as well as more neutrons which further cause fission of neighboring atoms. Hence, it is a chain reaction and it must be controlled, otherwise it may result in explosion. A nuclear reactor consists of fuel rods, control rods and moderator. A fuel rod contains small round fuel pallets (uranium pallets). Control rods are of cadmium which absorb neutrons. They are inserted into reactor and can be moved in or out to control the reaction. The moderator can be graphite rods or the coolant itself. Moderator slows down the neutrons before they bombard on the fuel rods.

Two types of nuclear reactors that are widely used –

- **Pressurized Water Reactor (PWR)**
- **Boiling Water Reactor (BWR)**

Component of NPP

- **Heat Exchanger**

In the heat exchanger, the primary coolant transfers heat to the secondary coolant (water). Thus water from the secondary loop is converted into steam. The primary system and secondary system are closed loop, and they are never allowed to mix up with each other. Thus, heat exchanger helps in keeping secondary system free from radioactive stuff. Heat exchanger is absent in boiling water reactors.

- **Steam Turbine**

Generated steam is passed through a steam turbine, which runs dueto pressure of the steam. As the steam is passed through the turbineblades, the pressure of steam gradually decreases and it expands in volume. The steam turbine is coupled to an alternator through a rotating shaft

- **Alternator**

The steam turbine rotates the shaft of an alternator thus generating electrical energy. Electrical output of the alternator is the delivered to a step up transformer to transfer it over distances.

- **Condenser**

The steam coming out of the turbine, after it has done its work, is then converted back into water in a condenser. The steam is cooled by passing it through a third cold water loop.

Advantages of NPP

- The nuclear power plant occupies less space in comparisons to thermal plants
- There are no fuel transportation, handling and storage charges since nuclear fuel requirement is less compared to coal.
- Less numbers of persons are required for its operation. Therefore, cost of operation is less.
- There is increased reliability of operations.
- No ash handling problem.
- Typically 90% efficient power generation capacity
- These plants are non-polluting unlike coal or oil power plant.
- Nuclear power generates lots of clean, stable energy.
- It lowers down the demand of coal and fossil fuels.

Disadvantages of NPP

- Capital cost is large
- Need skilled and trained manpower to operate.
- Problem of radioactive waste disposal.
- High degree of safety is needed.
- Radiation hazards to living and non living things.
- Pollution of land, water and air.
- Maintenance cost of plant is high.

Site Selection criteria

Important considerations in **site selection** of **nuclear power plants** (NPP) are

- **Availability of water supply.**

  Inside the reactor of the nuclear power plant, water is used as a moderator for cooling the heat generated by fission reaction and for other cooling purposes also so the requirement of water in a nuclear power plant is very high (that is twice the requirement of water in thermal power plant) so the plant must be located in the area where availability of water is plenty that is near the river, lake or by the seaside.

- **Distance from a populated area.**

  As we know that for fuel we use for the nuclear power plant is radioactive materials like uranium, thorium, etc., and though we take all the safety measures, we cannot make all the environment around the power plant radioactive free, so there should be a reasonable distance between the nuclear power plant and the nearest populated area from the safety point of view.

- **Nearness to the load center**

The transmission cost of electricity depends upon the nearness of the plant to the load center more the distance more the transmission cost; hence practically, a power plant should always be located near the load center, but we cannot place nuclear power plant that closes to the load center so for this authorities have to erect a nuclear power plant by taking all safety consideration.

- **Availability of space for disposal of waste**

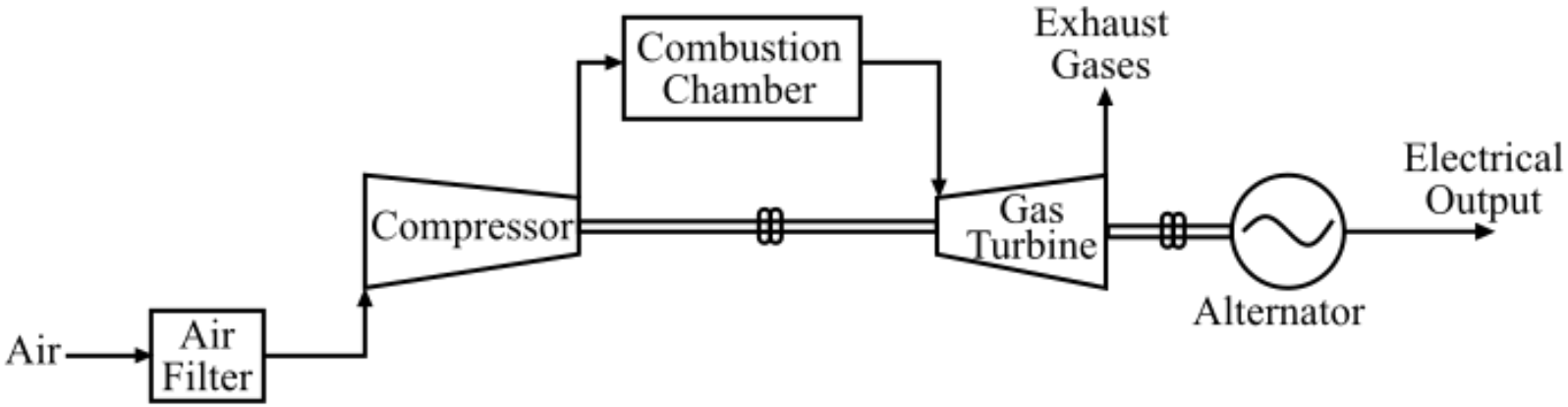
Like the other power plants, we cannot dispose of the wastage of nuclear power plant, after all the use of fuel to generate electricity. Even after decaying that fuel, it still remains radioactive and hazardous for us. So while selecting the location of the nuclear power plant, one must think of the availability of space for the disposal.

- **Accessibility**

There should be reasonable accessibility for plant personnel, haul equipment, and dispatch and receive heavily shielded radioactive materials.

Gas Turbine Power Plant

A power generating plant which has gas turbine as the prime mover for the generation of electrical energy, is called the gas turbine power plant. The block diagram of a typical gas turbine power station is shown in the figure.

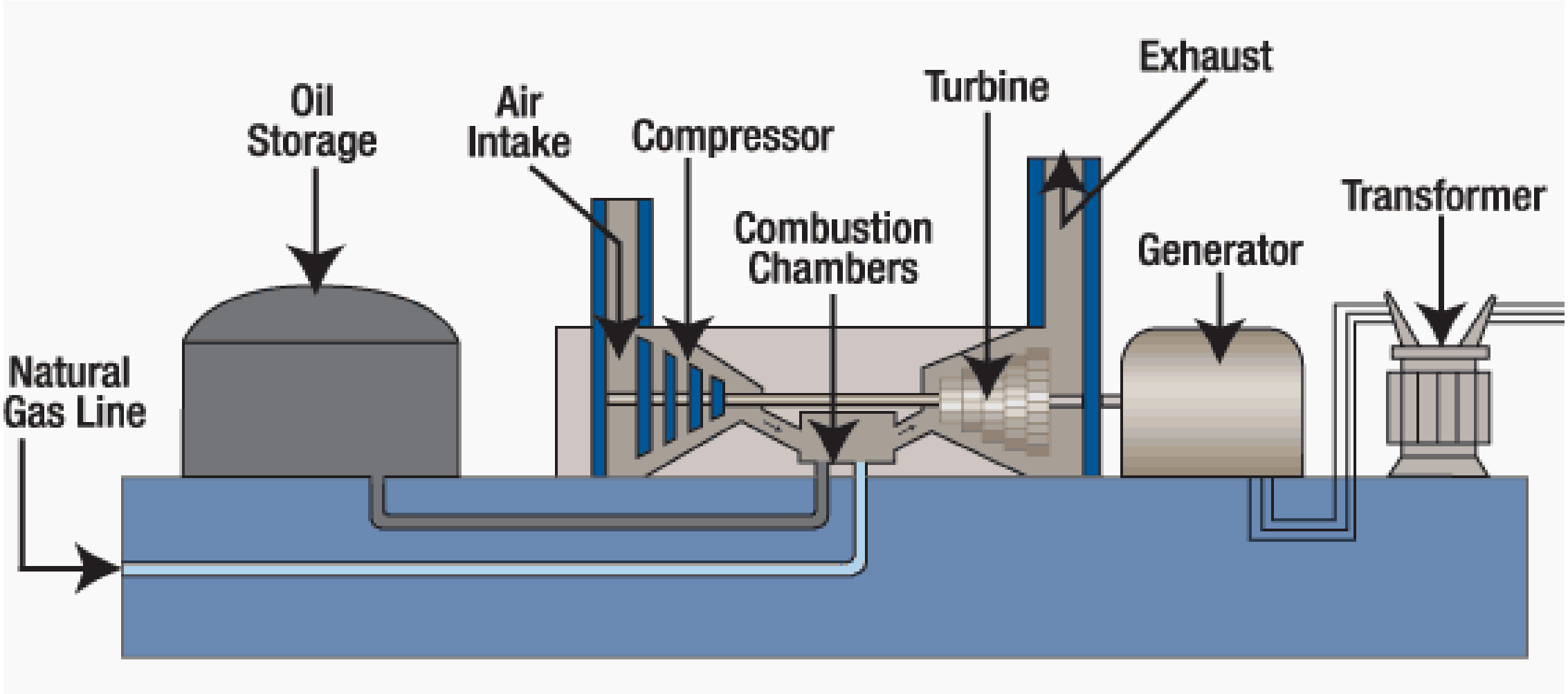


In a gas turbine power plant, natural air is used as the working agent. The air is compressed with the help of a compressor and fed to a combustion chamber where heat is added to this compressed

air, which increases the temperature of the air. The heat to the compressed air is added either by burning fuel in the combustion chamber or by the use of air heaters.

Then, the hot and compressed air from the combustion chamber is passed through the gas turbine where it expands and does the mechanical work, i.e. the heat energy of the compressed air is converted into mechanical energy. The gas turbine drives the alternator which converts the mechanical energy of the turbine into electrical energy.

Also, it is important to note that the compressor, gas turbine and the alternator are mounted on the same shaft so that the mechanical power of the turbine can be utilised for the operation of the compressor.


Oil
Storage
Air
Intake
Compressor
Turbine
Exhaust
Combustion
Chambers
Generator
Transformer
Natural
Gas Line

Components of a Gas Turbine Plant-

- **Inlet-**

  The inlet to the gas power plant is used to drive air in to the Gas turbine power plant. Only fresh air is driven into the gas turbine power plant. We can apply dry filters like paper foil and wet filters like paper Gas Turbine power plant consists of following parts. They are as follows:

- Exhaust
- Combustion Chamber
- Turbine
- Generator
- Compressor

- **Compressor-**

The compressor name itself defines the work done by it. It compressor the air to a greater extent. The purified air is sent to the compressor where it gets compressed to a very high pressure and is sent to the combustion chamber for combustion process to take place.

- **Combustion Chamber-**

  The compressed air from the compressor is sent to the combustion chamber. The process of combustion chamber is to do combustion. For combustion, we need a fuel and compressed air. So proper fuel will be mixed in the combustion chamber. The combustion process also depends on the type of fuel used and the quality of fuel used. The fuel may be injected in to the combustion chamber in the form of spray with the help of nozzles. Use of nozzles further increase the speed of fuel to combustion, which increases the efficiency of the combustion.

- **Turbine-**

In the compressor, the air is compressed. In the Combustion chamber, the air is further compressed, and heat energy is also added. So, this is transferred to the turbine blades with the help of nozzle which is a small opening. The efficiency of the turbine completely depends on the temperature and pressure of the gas coming out from combustion chamber.

- **Generator-**

  In the turbine, the energy of the gas is converted to mechanical energy. In the Generator the mechanical energy is converted to electrical energy. This is the only role of the generator in the Gas Turbine Power Plant. This Generator is coupled to the Turbine so the complete efficiency is obtained.

- **Exhaust-**

  The gases coming out from the turbine cannot be further used for combustion.In an inter connected power plant, it may be further used again. But in this Gas Turbine Power Plant, it is left to the open atmosphere far from the inlet.

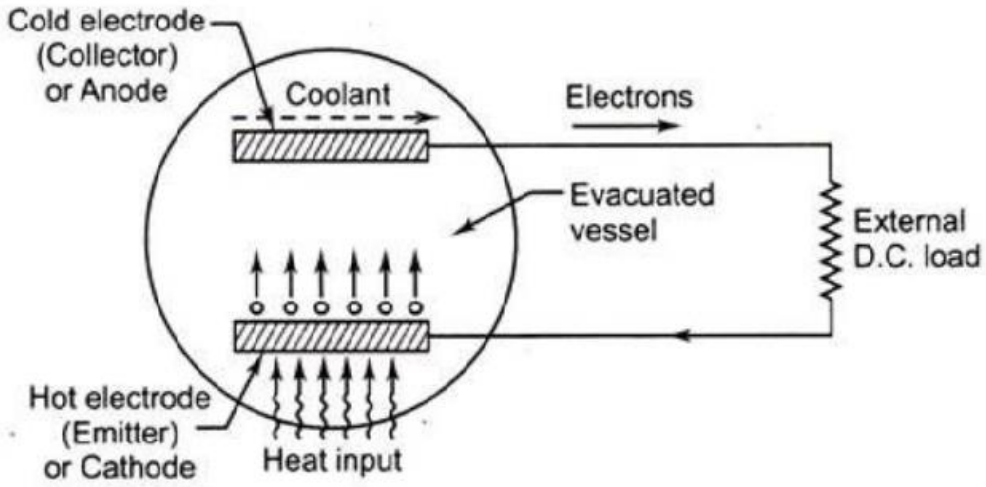
Advantages Of Gas Power Plant

- It is simpler in design because there are no boilers, no feed water supply, no cooling water.
- It can be used in region where large amount of water is not available for steam power plant.
- They are small in size, weigh less and have low initial cost per unit output.
- They are easy to install within short periods.
- They are quick-starting and smooth running.
- They offer flexibility by supplying electricity for power generation as well as by supplying compressed air for process needs.
- Water consumption is less compared to steam power plant.

Disadvantages of GPP

- An electric motor or an I.C. engine is necessary for starting the plant. The starting motor must bring the compressor well towards the operating speed So, starting is not simple as in the case of other power plants.
- Gas turbine plants have less vibrations when compared with reciprocating, engines of the same speed. However, the high frequency noise from the compressor is objectionable.
- High temperatures impose severe restriction on the servicing conditions of the plant.
- Overall efficiency is low since two-thirds of the total power output is used for driving the compressor.
- The blades of the turbine require special cooling methods due to the severity of operating temperatures and pressures. In practice, the temperatures at the entry of the turbine are as high as 1100°C - 1260°C. Hence, they should be made of special metals and alloys.

Thermionic Generator



Thermionic power converter, also called **thermionic generator, thermionic power generator** any of a class of devices that convert heat directly into electricity using thermionic emission rather than first changing it to some other form of energy.

The principle of thermionic converters is illustrated in Fig. In a thermionic converter two electrodes are placed in a container containing an ionised gas or cesium vapors to reduce the space charge. The cathode is heated by concentrating the rays on it. On heating of

cathode the electrons are emitted from it and travel to anode. The cathode and anode are connected externally through the load circuit.

The electrons return back to cathode through the external circuit and the current flows through the external circuit as shown in the figure. The heat energy is converted directly into electrical energy through a process similar to that in a steam plant where water is evaporated in a boiler and the steam is condensed after doing the useful work in an engine.

Thermoelectric Generator

- A **thermoelectric generator** (**TEG**), also called a **Seebeck generator**, is a solid state device that converts heat flux (temperature differences) directly into electrical energy through a phenomenon called the *Seebeck effect* (a form of thermoelectric effect). Thermoelectric generators function like heat engines, but are less bulky and have no moving parts. However, TEGs are typically more expensive and less efficient.

- Thermoelectricity refers to class of phenomena in which a temperature difference creates an electric potential or an electric potential creates a temperature difference.
- Thermoelectric generators could be used in power plants and factories to convert waste heat into additional electrical power and in automobiles as automotive thermoelectric generators (ATGs) to increase fuel efficiency.
- Thermoelectric generators have no moving parts which produce a more reliable device that does not require maintenance for long periods.
- When the junctions of two different metals are maintained at different temperature, the emf is produced in the circuit. This is know as seebeck effect.